

Module 03 Lab Exercise - Contribution Journal

ITAI 1371 - Introduction to Machine Learning

John Grubbs

For this lab, I contributed by working through the module_03_lab exercise code and experimenting with the provided win dataset. I reviewed how the dataset was organized into features and target classes, examined the class distribution and feature correlations, and worked through splitting the data into training and testing sets.

My main contribution to my own learning was taking time to understand what the provided code was actually doing rather than simply executing each cell. In particular, I investigated concepts such as correlation, random_state, and stratify, which helped me better understand how data is prepared for classification. I also completed the exercise categorizing different scenarios as supervised, unsupervised, or reinforcement learning.

Luis Adrian

My contribution to this lab was completing the different steps of the machine learning workflow and making sure the notebook worked correctly. I worked with the Wine dataset by exploring the data, checking the features and missing values, and reviewing the class distribution.

And I also compare the performance of the two models of the lab and analyse the result. The model of "Logistic Regression" achieved 88.9% accuracy, while the model of the "Decision Tree" achieved 83.3%. And I review the classification report and confusion matrix to better understand the predictions and errors.

Keira Burdine

My contribution to Module 03 focused on completing the machine learning lab exercises, experimenting with model features, evaluating the results, and connecting the concepts to a real-world cybersecurity application. I worked through the notebook cells and reviewed the outputs as I progressed so I could understand what each section was demonstrating instead of only running the code. This module gave me hands-on experience with supervised, unsupervised, and reinforcement learning and helped me see how each type can be used for different problems.

During the wine classification portion of the lab, I worked with the available features and tested how feature selection affected the logistic regression model. My first combination used alcohol, color intensity, and proline. That model produced an accuracy of 83.3%, compared with the original model accuracy of 88.9%. Because the first combination did not improve the model, I tried another set of features. I selected color intensity, OD280/OD315 of diluted wines, and proline. This combination produced 100% accuracy in the lab's test split and improved on the

original result. My contribution included running these experiments, comparing the accuracy results, and using the output to understand why feature selection matters in machine learning.

I also completed the machine learning type assessment by identifying the appropriate learning method for five different scenarios. I identified house-price prediction and spam classification as supervised learning, customer grouping and hidden-topic discovery as unsupervised learning, and teaching a robot to play chess through repeated games as reinforcement learning. I received a score of 5 out of 5 on this section. This activity helped me practice recognizing the difference between labeled data, pattern discovery without predefined labels, and learning through feedback.

For the real-world application portion, I connected the assignment to cybersecurity. I described a supervised learning system that could help detect potential cyberattacks using network traffic, login activity, IP addresses, timestamps, and examples of normal and malicious behavior. This was an

important part of my contribution because it allowed me to apply the machine learning concepts from the lab to an area that interests me professionally. It also helped me understand how AI and cybersecurity can work together to identify suspicious activity.

Overall, I contributed by completing the assigned lab activities, testing and comparing different model features, evaluating model accuracy, completing the machine learning type assessment, and developing a cybersecurity application example. I also completed the reflection and reviewed the machine learning lifecycle from problem definition and data collection through model training, evaluation, deployment, monitoring, and continuous improvement. My biggest takeaway from my contribution is that machine learning requires both technical steps and analysis. Running a model is only one part of the process; understanding the data, selecting useful features, evaluating the results, and deciding how the model could be applied are equally important.